

Course Plan and Policies

CL-202: Chemical Engineering Thermodynamics

Semester-I, 2026-2027

Indian Institute of Technology Gandhinagar

Instructor	Teaching Assistant
Kaustubh Rane Office: Academic Block 5, Room 321 Email: kaustubhrane@iitgn.ac.in	TBA

Class Location and Meeting Times

Lectures/Tutorials: TBA

Extra Classes: TBA

Mode of Engagement: In-person

Reference Texts

- J. M. Smith, H. C. Van Ness, M. M. Abbott, M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics*, 8th edition, McGraw-Hill Education, 2018. [Primary Textbook]
- J. M. Prausnitz, R. N. Lichtenthaler, E. C. de Azevedo, *Molecular Thermodynamics of Fluid- Phase Equilibria*, 3rd edition, Prentice Hall, 1999.

Course Objectives

The aim of the course is to enable the students to develop understanding of thermodynamics involved in unit operations and harness their knowledge to design chemical engineering processes.

Expected Outcomes

At the end of this course the students should be able to understand the underlying thermodynamics of the unit operations in chemical engineering and shall be able to carry out thermodynamic analyses of a variety of chemical processes that cut across traditional departmental boundaries.

Learning Objectives

The course is designed to achieve following learning objectives:

- Students will learn to calculate the thermodynamic properties of pure components and mixtures.
- Students will learn the fundamentals of solution thermodynamics and phase equilibria (mainly Vapor-liquid equilibrium) in binary and multicomponent systems along with reaction equilibrium.

Grading (To be finalized)

The final grade shall be awarded based on the total score of the student that will be calculated using the following weights.

- **Quizzes:** TBA
- **Assignments/Tutorials:** TBA
- **Mid semester exam:** TBA
- **End semester exam:** TBA

CL-202: Chemical Engineering Thermodynamics Syllabus

Instructor: Kaustubh Rane

Lectures/Tutorials: TBA

Introductory Concepts

- Use of Jacobians in thermodynamics

Thermodynamic Properties of fluids

- Property relations for homogenous phases
- Residual properties
- Two-phase systems
- Thermodynamic diagrams

Systems of variable composition: Ideal behavior

- Fundamental property relation
- Chemical equilibrium as a criterion of phase equilibrium
- The ideal-gas mixture
- The ideal solution
- Raoult's law

Systems of variable composition: Nonideal behavior

- Partial properties
- Fugacity and fugacity coefficients
- Fugacity and fugacity coefficients for species i in solution
- Excess Gibbs energy
- Activity coefficients from VLE data

Phase equilibria at low to moderate pressures

- The Phase Rule. Duhem's theorem
- Phase behavior for vapor/liquid systems
- Low pressure VLE from correlations of data
- Composition dependence of fugacity
- Henry's law

Solution thermodynamics

- Relations among partial properties for constant-composition solutions
- The fundamental residual property relation
- The fundamental excess property relation
- Evaluation of partial properties
- Property changes of mixing
- Systems of limited phase miscibility

Chemical reaction equilibria

- The reaction coordinate
- Application of equilibrium criteria to chemical reactions
- The standard Gibbs energy change and equilibrium constant
- Effect of temperature on equilibrium constant
- Evaluation of equilibrium constants
- Relations between equilibrium constants and composition
- Calculation of equilibrium conversions for single reactions
- Phase rule and Duhem's theorem for reacting systems
- Multireaction equilibria